Natural Language Processing
Final Project

English–Arabic Code-Switching Normalization to Arabic Using Sequence-to-Sequence Models

Project Overview

Code-switching is a common linguistic phenomenon where speakers alternate between two languages within a sentence or discourse. In Arabic social media, chat messages, and informal communication, English–Arabic code-switching is extremely frequent:

أنا today مشغول في meeting

The goal of this project is to build a neural Sequence-to-Sequence (Seq2Seq) model that converts English–Arabic code-switched text into fully Arabic text, while preserving the original meaning.

Example:
Input (Code-Switched): أنا today مشغول في meeting
Output (Arabic-Only): أنا اليوم مشغول في اجتماع

This task is framed as a **text-to-text generation problem**, where the model learns to:

- Translate English words into Arabic
- Preserve Arabic grammatical structure
- Handle informal and noisy text
- Maintain semantic coherence

Learning Objectives

By completing this project, students will be able to:

1. Understand the linguistic phenomenon of **code-switching**
2. Formulate normalization and translation as a **Seq2Seq task**
3. Build encoder–decoder models using **LSTM or GRU**
4. Apply **attention mechanisms** for mixed-language inputs
5. Preprocess noisy multilingual text
6. Evaluate generation quality using standard NLP metrics

English–Arabic Code-Switching Datasets

Students may use publicly available datasets specifically designed for Arabic–English code-switching.

1- CALCS – Corpus of Arabic–English Code-Switching

A high-quality dataset containing Arabic–English code-switched sentences annotated for language identification. (t.ly/2T3PQ)

2- Arabic-English-Intra-Word-Code-Switching-Corpus

A social-media-based corpus containing naturally occurring Arabic–English mixed tweets. (t.ly/Twmqg)

3- https://huggingface.co/datasets/SDAIANCAI/Ar-En-Code-Switching-Textual-Dataset/viewer/default/train?row=1

4- https://www.kaggle.com/datasets/sdaiancai/arabic-english-code-switching-textual-dataset?select=README.md

5- https://huggingface.co/MohamedRashad/Arabic-Whisper-CodeSwitching-Edition

Students are encouraged to generate synthetic code-switched data by:

- Replacing Arabic words with English equivalents
- Injecting English terms into Arabic sentences
- Pairing mixed input with clean Arabic output

Model Architecture

Students must build a Seq2Seq Encoder–Decoder model with an attention mechanism, where the encoder is implemented using a Bidirectional LSTM or GRU to encode the mixed-language input sequence, and the decoder is implemented using an LSTM or GRU to generate an Arabic-only output sequence. The model should incorporate an attention mechanism to effectively align English words in the input with their corresponding Arabic translations and improve translation quality.

Evaluation Strategy

Benchmarking will be conducted by comparing the proposed model against well-established baseline models to objectively evaluate its performance and demonstrate its added value. Baselines may include a standard Seq2Seq encoder–decoder model without attention, a unidirectional RNN-based architecture, and Transformer based approach where applicable. All models will be trained and evaluated under identical experimental settings, including the same training–validation–test splits, preprocessing pipeline, and hyperparameter tuning protocol to ensure fairness. Performance will be measured using appropriate evaluation metrics such as BLEU score, ROUGE, Perplexity, accuracy, precision, recall, F1-score, or sequence-level exact match, depending on the task. Statistical significance testing may also be applied to verify whether performance differences are meaningful.

Quantitative Metrics

Metric	Purpose
BLEU	Measures translation quality
ROUGE	Measures content overlap
Word Error Rate (WER)	Measures correction effort
Token Accuracy	Measures generation correctness

Qualitative Evaluation

- Manual inspection of outputs
- Error analysis on:
    - Incorrect translations
    - Grammar mismatches
    - Missing words

Expected Deliverables

- Complete Research Paper (Latex Springer Format on Overleaf) and should cover the following components
    - Problem definition
    - Dataset description
    - Preprocessing steps
    - Model architecture
    - Evaluation results
    - Error analysis
- Source Code (Uploaded on GitHub and share it to me)
- Presentation: 10–15 slides summarizing the project

Grading Rubric:

Task	Expected Due Date	Weight
Qualification assessment1	Week 3	5 pts
Data Collection, Text cleaning and processing using regular expressions	Week 5	3 pts
Design NLP Model	Week 7	4 pts
Building, training and testing the NLP Model	Week 10	6 pts
Qualification assessment2	Week 11	5 pts
Frontend- Building User interface using Streamlit or Gradio Final Project Submission	Week 12	2 pts
Discussion	Week 13	5 pts

Use of AI Tools

Students may collaborate with classmates and use Generative AI (Assistive Use) while studying the course, as long as they avoid direct plagiarism as outlined below. Students are allowed to seek assistance using artificial intelligence tools for supportive, non-essential purposes, such as brainstorming ideas, generating suggestions, proposing preliminary or general ideas or titles, linguistic proofreading, rephrasing and reformulation, analyzing results, and accessing sources and references. This does not include drawing conclusions. However, it is not allowed to copy and include the content directly in the final submitted work. The permitted percentage of AI-generated content (Assistive Use) must not exceed **20%** of the total submitted academic, code or research content.

Plagiarism

Using a tool does not transfer authorship. If you use any tool to generate code, you are not considered the author of that code. You cannot submit code generated by another person/tool and present it as your own. This constitutes serious plagiarism and will be reported to the Dean.